Neoantimon2 — variants to traceable candidates

Native analysis and a separate External route; references and predictors are obtained separately

Patients + samples
Explicit identity mapping
Per-sample VCF inputs

Reference bundle
Genome + annotation
Transcript / protein

Patient-level HLA
User supplied
Applied to each sample

Optional local predictors
Class I: NetMHCpan / MHCflurry
Class II: NetMHCIIpan

Optional evidence
RNA / DNA / MS

A NATIVE

Variant events
Normalize
REF / ALT

→ **Origins + translation**
Multiple paths
per event

→ **Peptide sequences**
Shared peptides
Origins retained

→ **Peptide–HLA candidates**
Sample + patient
and presenter

→ **Prediction + evidence**
Local predictors
Observations

→ **Prioritize + explain**
Rank / eligible grade
Traceable results

Multiple paths can share a peptide; every candidate–origin link is retained.

Optional: export after the native project completes

B EXTERNAL — a separate derived view

Export bundle
Queries + all origins
Completed project

→ **Model outside Neoantimon2**
Run separately
User-selected model

→ **Standard format**
Direct standard TSV
or explicit converter

→ **Strict matching**
Peptide or
peptide + HLA

→ **External results + explanations**
Derived analysis
Native unchanged

Import requires the original completed project. Strict matching preserves all origin links.
External grades require MHC percentiles + eligible core somatic evidence; raw scores alone do not grade.

Specialized workflows — explicit opt-ins; external scores do not feed these views

Longitudinal
Completed sample runs + settings
Native grades / ranks retained

Vaccine-set selection
Native evidence + explicit policy
Native grades / ranks retained

Restricted indirect-HLA / allo
Own inputs + restricted profile
Not an official PIRCHE implementation

Missing evidence ≠ zero ≠ evaluated negative • Research outputs, not clinical validation